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Soil health management strategies for climate-resilient agriculture

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Abstract

Achieving global agricultural production goals is becoming more difficult due to multiple challenges and increasing climate variability. In response, sustainable agriculture focuses on maintaining soil health worldwide, acknowledging the vital role of soil microorganisms in preserving soil quality and productivity. Plant growth-promoting rhizobacteria (PGPR), arbuscular mycorrhizal fungi (AMF), cyanobacteria, and beneficial nematodes are among the diverse organisms that support nutrient transformation, phytohormone production, plant resilience against abiotic stresses, and nutrient cycling within the soil system. Regenerative agriculture (RA) and organic farming strategies are essential in boosting soil microfauna diversity and abundance. Under the soil security framework, crop diversification, intercropping, and crop rotation are seen as effective strategies. However, achieving long-term adaptation requires updating agricultural production methods to current climate conditions, which calls for comprehensive, system-wide actions and significant policy and institutional changes at both national and local levels. Establishing beneficial relationships between plants and microbes, using advantageous microbial agents, appears to be a promising strategy for optimizing agricultural land use in the face of climate change.

Keywords Agroforestry system, Conservation agriculture, Ecosystem sustainability, Regenerative agriculture, Soil fertility

1 Introduction

Global climate change is widely acknowledged as a major threat to vegetation growth and agricultural sustainability [73]. As the global population is expected to reach 8.9 billion by 2050, agricultural systems, especially in developing countries, will face growing pressure to produce more food sustainably [170]. It is estimated that the global cereal equivalent (CE) food demand will increase from 10,094 million tons in 2030 to 14,886 million tons by 2050 [113]. Meanwhile, malnutrition remains a problem: in 2020, 149.2 million children under five were stunted, 45.4 million wasted, and 38.9 million overweight [301]. Environmental degradation worsens these issues, with 2 billion hectares,

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25% of the world's land, already degraded and 24 billion tons of productive soil lost each year [88]. According to the IPCC [111], climate change is expected to raise global temperatures by 1.5 °C by 2050 and between 2 and 4 °C by 2100. This increase is likely to cause notable impacts, including temperature fluctuations, shifts in rainfall patterns, rising sea levels, freshwater shortages in coastal areas, and more frequent extreme weather events [71]. For example, climate-driven drought in eastern India has already begun to reduce regional crop yields. During the severe 2022 drought in Bihar, rice production dropped roughly 23% [196]. Additionally, in India, rising temperatures, altered rainfall patterns, and increasing CO2 levels are altering the incidence, severity, and geographic spread of major crop pests and diseases [284]. Water demand is projected to increase by 20–30% by 2050 [40], and economic losses due to floods (32%) and droughts (7%) have been significant between 1970 and 2023 (STATISTA, [274]).

Crop productivity is becoming more vulnerable, with heat stress expected to reduce soybean, maize, and wheat yields by about 19%, 22%, and 24%, respectively [65]. Between 1990 and 2015, roughly 3.1 billion tons of carbon were lost from 2 million km^2 of land, averaging approximately 60 metric tons per km^2 per year [237]. Climate impacts vary greatly by region and crop, with central India experiencing larger fluctuations in rice yields, which increases South Asia's overall vulnerability [209, 242]. Meanwhile, excessive use of synthetic fertilizers and pesticides has caused degradation of agroecosystems and water contamination, threatening aquatic life and human health [66]. In response, sustainable adaptation strategies are being developed globally [54]. Sustainable agriculture combines biological, chemical, physical, and ecological principles to maintain productivity while protecting environmental quality [2]. Although maximizing yields is essential, farming systems must equally deliver ecological, social, and economic advantages. Low-yielding areas present opportunities for targeted productivity improvements, achievable through organic, conventional, or hybrid approaches [74].

In this context, scientists play a crucial role in translating soil health research into practical tools for sustainable land management. Sharing their expertise enables growers to adopt evidence-based practices. Regenerative Agriculture (RA) is one approach that enhances soil health, biodiversity, and watershed function while improving overall ecosystem services [280]. It also aims to reduce atmospheric carbon accumulation by increasing carbon sequestration in soils and plants, complementing other sustainable practices such as climate-smart agriculture and conservation agriculture [283]. The main philosophy of RA is to do more with less land, fewer chemical inputs, reduced water use, lower GHG emissions, and less degradation [162].

A key part of sustainable land management is assessing and protecting soil health, which includes the biological, chemical, and physical qualities that support productivity and vital ecosystem functions. It is affected by both natural and human-made factors, influencing agriculture, biodiversity, and recreational areas. Important indicators include microbial diversity, stability, activity, and abundance, with soil microbes, such as bacteria and fungi, playing vital roles in nutrient cycling and plant growth (Vander [3, 148, 167, 290]). Research over seven years has shown that organic farming increases soil microbial carbon compared to conventional methods [62], while practices such as tillage and moisture conservation also significantly impact soil health and structure [137, 245]. To expand regenerative and climate-smart practices, understanding local conditions and empowering farmers are essential. Social protection programs, supportive policies,

institutional frameworks, funding mechanisms, and risk management strategies are all necessary to build climate resilience in agriculture [78]. The mutually beneficial relationships between soil microbes and plants, including mycorrhizae and PGPR, are crucial for improving nutrient uptake under stressful conditions [19, 151, 263–265]. Still, ongoing cultivation and unsustainable land use continue to harm soil health [294], emphasizing the urgent need to restore or protect it, especially in the face of climate uncertainty.

Exploring beneficial plant–microbe interactions and stress-tolerant microbial strains offers great potential for climate-resilient farming. Sustainable crop production must now consider both past and emerging climate variability [154]. Institutional support and effective strategies are essential for addressing climate risks and promoting agricultural sustainability. This review examines the role of beneficial microbes in improving soil health, identifies best practices in soil management, and proposes soil health indicators to expand climate-smart agriculture for sustainable development.

2 Methodology

This systematic review followed the PRISMA framework to identify, screen, and synthesize evidence on soil health management strategies under changing climate conditions (Fig. 1). A comprehensive literature search was conducted across four major academic databases, Scopus, Web of Science, ScienceDirect, and Google Scholar, which yielded 2977 peer-reviewed records. An additional 1957 documents were retrieved from institutional and organizational repositories, including FAO, UNEP, and ICAR, to ensure the inclusion of grey literature and region-specific reports. The search strategy used combinations of keywords, including soil health, soil quality, soil carbon, regenerative agriculture, climate-smart agriculture, land degradation, soil microbial communities, and climate change adaptation. All retrieved records were imported into a reference management system, and duplicates were removed. After deduplication, 2007 records from databases and 1147 records from institutional sources remained for initial screening. Titles and abstracts were screened for relevance, scientific rigor, and thematic alignment,

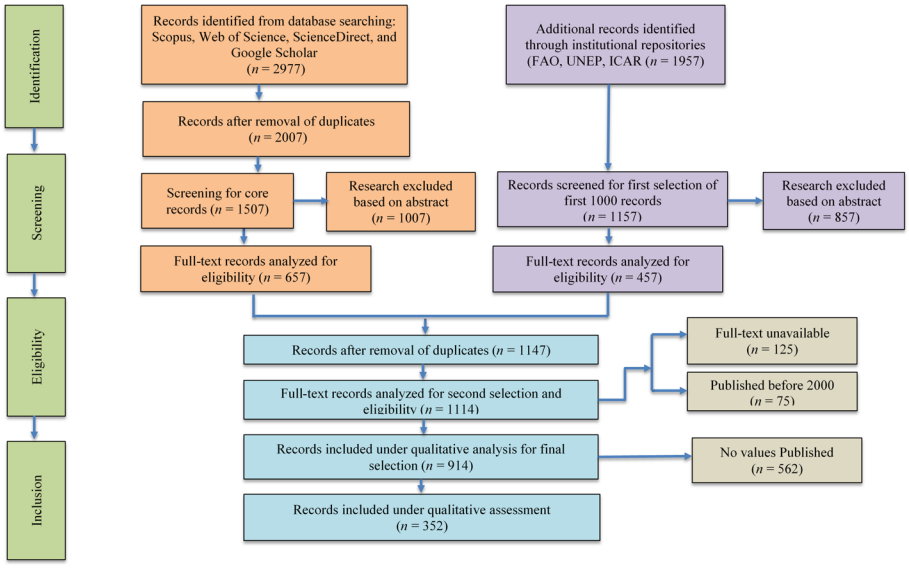


Fig. 1 Search criteria and methodological framework considered under the systematic literature review through a modified version of the PRISMA "preferred reporting items for systematic reviews and meta-analyses" guidelines and rules. (Adapted from Moher et al. 2010; Edrisi et al. 2022).

resulting in the exclusion of 1007 database-derived records and 857 institutional records. Full texts of the remaining studies were assessed in two phases. In the first phase, 657 peer-reviewed articles and 457 institutional documents were evaluated for eligibility. After removing an additional 1147 duplicates, 1114 full-text records proceeded to detailed assessment. Studies were excluded if full texts were unavailable, published before 2000, or lacked quantitative or qualitative soil health indicators; this accounted for 125, 75, and 562 exclusions, respectively. Studies were included if they focused on soil health, biological, chemical, or physical soil indicators, climate-smart soil management, or soil-based climate resilience; presented empirical, modelling, experimental, or meta-analysis data; and reported measurable soil outcomes related to carbon dynamics, nutrient cycling, structural improvements, or ecosystem services. Studies were excluded if they lacked empirical evidence, focused on non-agricultural soils, lacked relevant soil metrics, or fell outside the thematic scope. After applying all inclusion and exclusion criteria, 914 studies were deemed suitable for qualitative assessment, and 352 high-quality studies were included in the final synthesis. These studies formed the evidence base for evaluating soil process dynamics, microbial functioning, carbon sequestration potential, and climate-resilient soil management practices.

3 Soil biodiversity and ecosystem sustainability

Soil, though occupying only a small portion of Earth's surface and containing a limited amount of organic matter, harbors an extraordinary diversity of life, particularly in its topsoil. Notably, more than 25% of all terrestrial life exists within the soil, highlighting its critical role in sustaining food production, mitigating pollution, supporting aquatic and terrestrial ecosystems, enabling nutrient cycling and retention, and regulating the climate [199]. However, soil biodiversity faces escalating threats from land degradation, pollution, urbanization, mismanagement, and climate change—pressures that mirror those impacting broader natural resources. Among soil's biological components, plant roots and microorganisms are key drivers of ecosystem functioning, with microbes accounting for 60–80% of soil activity, including nutrient cycling and decomposition of organic matter [44]. A sustainable soil ecosystem, essential for robust crop growth, relies on the complex interplay between soil microorganisms, structural properties, and plant systems. Gaining deeper insight into microbial communication and functionality is crucial to prevent adverse outcomes associated with poor soil management [250, 275].

Due to its immense biological diversity, soil is considered one of the largest reservoirs of biodiversity on Earth [20]. This biodiversity plays a fundamental role in ecosystem sustainability—providing habitats, regulating climate, preserving water quality, controlling pollution, and enhancing food security [26, 220, 275]. Soil biota also boosts ecological stability by influencing plant diversity, productivity, and species interactions, thereby improving ecosystem resilience [20, 275]. In particular, root-associated microorganisms help plants tolerate climate-induced stresses such as drought and temperature extremes. However, soil biodiversity is increasingly at risk in densely populated and intensively cultivated regions, where biodiversity loss, overgrazing, chemical pollution, and land degradation are major concerns [26]. Protecting soil biota is therefore essential for achieving sustainable agricultural goals and requires proactive conservation strategies [275].

Importantly, the value of soil biodiversity extend far beyond academic interest; it is increasingly utilize to address critical local, regional, and global issues, including

biodiversity loss. Recognizing soil as a dynamic, living system encourages the development of integrated, cross-sectoral solutions. Both microscopic and macroscopic soil organisms drive vital ecosystem processes by cycling nutrients and energy, thereby supporting plant and animal life across terrestrial ecosystems. Nature-based solutions rooted in soil biodiversity, such as conserving natural landscapes, rehabilitating degraded land, promoting sustainable farming practices, and reimagining urban spaces for human and ecological benefits, provide powerful ways to address climate change, biodiversity loss, and sustainability [21]. As we aim for a sustainable future, managing this essential and diverse resource beneath our feet is crucial.

4 Soil health components for sustainable agriculture

4.1 Distribution of soil microorganisms

Healthy soil sustains a diverse community of organisms, including vertebrates, earthworms, mites, insects, nematodes, fungi, bacteria, and actinomycetes [295]. These organisms interact closely, forming a complex web within the soil ecosystem. While soil fungi may exceed 100,000 species, only a fraction, ranging from 18,000 to 35,000, are formally described. Nematodes and mites represent about 3% and 5% of total species, respectively [35]. Estimating bacterial and archaeal species is challenging due to classification inconsistencies and cultivation difficulties [55]. Soil's diverse organisms greatly enhance its biodiversity, interacting intricately with plants. Soils support a wide array of microorganisms and larger organisms, including plants and fauna, provided that a sufficient carbon source is available to fuel their activity. Bacteria are plentiful but have a small biomass due to their size, while actinomycetes have comparable biomass but are fewer and larger. Fungi, though fewer in number, contribute significantly to overall soil biomass [259]. In tilled soils, bacteria, actinomycetes, and protozoa dominate due to greater tolerance to soil disturbance [75, 300]. Soil total biomass generally falls within the range of 8–15 t ha^{-1}, with variability driven by soil type, organic matter content, and land-use practices [116]. Nitrogen-fixing microorganisms, such as cyanobacteria and *Rhizobium*, play a vital role in enhancing soil fertility, as illustrated in Fig. 2 [136]. The rhizosphere around active roots houses over 30,000 bacterial species, indicating a positive link between organic compounds and microorganism population [289]. Organic-rich soils support more microflora and fauna than those with minimal organic matter,

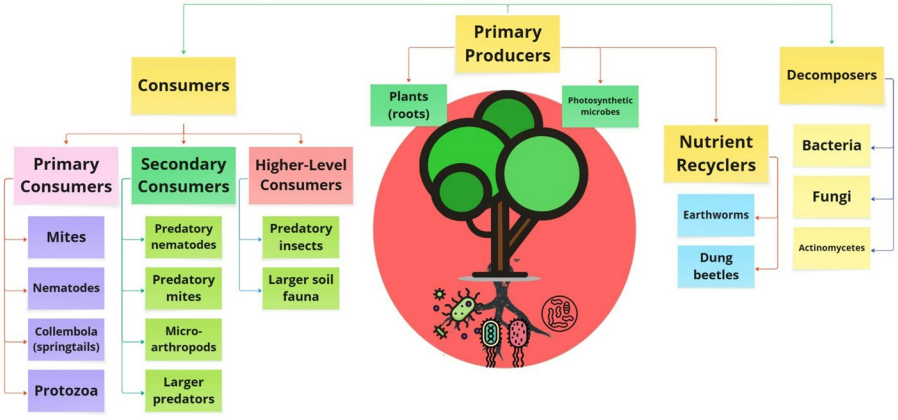


Fig. 2 A schematic soil food chain showing some of the major soil biota groupings

especially in forest soils known as "living soil" due to their rich organism concentration [116].

4.2 Farming practices to improve soil health

Ensuring soil well-being is fundamental to achieving agricultural productivity and sustainability. Soil health extends beyond crop yields to influence food safety, human and animal health, and water quality [224]. Effective management requires integrating physical, chemical, and biological indicators to assess the impact of farming practices and climate on soil functions, especially nutrient and carbon cycling. Even minor changes in soil organic carbon (SOC) per unit area can significantly affect the global carbon balance. Enhanced SOC stability improves soil structure, water and nutrient retention, buffering capacity, and microbial activity [159].

To support sustainable farming, the [289] outlines four core soil health principles: minimize disturbance, maximize soil cover, enhance biodiversity, and maintain living roots. These guidelines serve as a practical framework for improving soil function. Crop residue management, for example, can increase SOC, whereas certain cultivation practices may reduce SOC by accelerating the breakdown of physically protected carbon [23, 121]. Various conservation practices relevant to this are summarized in Table 1.

4.2.1 Maximize soil cover

Four essential farming techniques that play a key role in preserving soil cover and improving soil health are: (1) cover cropping, (2) mulching, (3) residue retention, and (4) green manuring. One such technique, cover cropping, involves growing crops with canopies that intercept rainfall and reduce soil erosion [127]. Cover crops, especially legumes, efficiently fix atmospheric nitrogen by partnering with rhizobium bacteria, making nitrogen available to plants. For example, cowpea can fix 25–30 kg ha^{-1} yr^{-1} of nitrogen [266]. Non-legumes, such as mustard, can serve as cover crops, too, inhibiting soil-borne pests with compounds like glucosinolates [108]. Mulching is another important practice that reduces water loss, controls erosion, and maintains favourable soil temperature and moisture conditions [130]. Residue retention, which involves leaving crop residues on the soil surface, protects the soil, supports microbial activity, and enhances soil moisture and carbon sequestration [152]. In general, the application of NPK and NPK + FYM results in increased biomass production and thus carbon input into the soil by 3–10 and 5–20 times, respectively, compared with no fertilizer addition [122]. Crop biomass production is negatively affected by poor nutrient availability under continuous cropping, thereby lowering productivity [262]. In a central Indian vertisol, Lakaria et al. [159] found that carbon addition to the soil increased from 228 to 332, 369 to 539, 44 to 65, and 256 to 374 kg ha^{-1} yr^{-1} through soybean leaf fall, roots, nodules, and rhizodeposition. When soybeans were treated with a 50% soil test crop response (STCR) dose plus a 5 t FYM treatment, the maximum C addition was observed. Similarly, the contribution of carbon from wheat crop through stubbles, roots, and rhizo-deposition has been observed to be 1033, 131, and 1446 kg ha^{-1} under the 50% STCR dose + 5 t FYM application. Finally, green manuring, which involves incorporating whole plants into the soil, improves organic matter content, enhances water retention, and stimulates microbial activity [180]. Collectively, these practices strengthen soil health and contribute to the long-term sustainability of agricultural systems.

Table 1 Effect of different conservation practices on soil health

Intervention	Crop/cropping system	Impact	References
Mulching	Green chilli	Increase the yield to 5 t ha^{-1} due to application of FYM as mulch (5 t ha^{-1})	Kumar et al. [152]
	Sorghum	In-situ surface mulching of Sunn hemp with deep tillage decrease the runoff and increase the soil profile moisture.	Narayan et al. [212]
	Wheat	IAA production, phosphate solubilization, and siderophores due to *Bacillus spp*. and *Pseudomonas spp*.	Joshi and Bhatt, [138]
	Rye	Acts as an allelopathic cover crop and has weed suppressing ability	Singhal et al. [266]
Sustainable Crop Rotation	Rice-wheat-summer mung bean	DHA activity was higher at maximum tillering (48.4%), flowering (8.6%), and after harvesting (19.1%) in rice-based CSA systems over maize-based conservation agriculture systems in rhizosphere soil.	(Jat et al. [117]
	Maize-wheat	Soil and moisture conservation.	Kumar et al. [152]
	Rice-wheat	Due to Integrated nutrient management, soil organic carbon increased to 6.80% as compared to conventional (5.10%).	Yaduvanshi, [310]
	DSR-ZT wheat	The grain yield of ZT wheat (5.03–6.90 t ha^{-1}) in sequence with DSR was 0.3–0.6 t ha^{-1} higher than what was attained after puddled rice systems.	Yadav et al. [308]
	Wheat-Fallow-Field pea	Provide enough cover to prevent soil deterioration and higher MBC found in the wheat-pea cover crop rotation.	'O'Dea et al. [218]
Diversified Cropping System	Castor + Green gram	Minimize the runoff & soil erosion loss.	Kumar et al. [152]
	Rice-wheat-rice-chickpea	Increase soil microbial biomass carbon by adoption of INM to 241.2 kg m^3 as compared to conventional (167.3 kg m^3).	Borase et al. [39]
	Pigeon pea+ Green gram (1:6)-Sesame	Fix atmospheric nitrogen in soil	Kumari et al., [157]
	Maize-Onion	High system yield was obtained 12.3 t ha^{-1}	Gangwar et al. [85]
Intercropping	Maize + Pigeon pea	Increased microbial density	Joshi et al. [136]
	Maize + Soybean	Increase the grain yield	Mudare et al. [202]
	Rice + Brahmi	More suppresses the weeds than a sole crop	Joshi et al. [135]
	Maize + Lablab	Increase infiltration and reduce surface runoff	Punyalue et al. [238]
	Maize + Beans + Squash	Squash acts as a ground cover by reducing weed density and the maize + bean intercropping system ensured optimum utilization of light	Postma and Lynch [236]
Minimum disturbance	Maize-Wheat	Reduced the soil compaction to a great extent	Ghuman and Sur, [90]
	Maize	Decrease the bulk density to 1.44 Mg m^{-3}, compared with conventional practices (1.54 Mg m^{-3})	Kannan et al. [142]
	Maize-wheat	Increase soil organic carbon to 4.5% using INM, up from 3.67% in the conventional system	Hazra et al. [99]
	Wheat	More residue retention increased the soil water content	Salahin et al. [252]
	Sugarcane	Increase the nitrogen content by inoculation with *A. diazotrophicus*	Joshi et al. [136]

Table 1 (continued)

Intervention	Crop/cropping system	Impact	References
Agroforestry	Cowpea + sapota	Improved carbon sequestration (9.73 Mg CO_2-equivalent ha^{-1})	Jinger et al. [129]
	Subabul-Gliricidia	Increase in organic carbon of 1.7–2.3 times compared to a control	Kaur et al., [144]
	Gliricidia + trench and Subabul + trench method.	Decrease the soil loss and increase the soil organic carbon by 44%	Hombegowda et al. [103]
	Sapota-based agri-horticulture system	With bench terrace and trenches, the runoff was reduced by 16–34% and soil loss by 15–25%.	Kumar et al. [153]
	Desert grass and Anjan grass	Stabilize the sand dunes due to deeper roots	Chauhan, [50]

4.2.2 Maximize biodiversity

Diversified cropping systems, integrated farming, agroforestry, and the use of organic manure/compost are potent methods for boosting biodiversity and soil health. Crop rotations diversify food sources for soil microbes, curbing pathogens and aiding nitrogen fixation [120]. Integrating agriculture with dairy, poultry, and other activities enriches soil fertility and cuts erosion risks [134]. These practices also create habitats for pollinators and soil organisms, fostering disease resistance and plant growth [289]. Agroforestry enhances biodiversity and soil quality, providing various ecosystem services such as moisture retention and nutrient availability [51, 77, 128, 129]. Additionally, organic manure application maintains soil fertility, with vermicompost providing additional benefits due to its nutrient-richness and support for microflora [69]. The sustainability and productivity of agroecosystems depend on the preservation of important microbial processes. The distribution of microbial functional groups, which drive processes such as soil carbon storage and nutrient turnover, is shaped by management practices. In contrast, soil aggregates provide essential habitats that support their activity. According to Helgason et al. [100], the main ways to affect soil microbial populations are by changing the soil's structure or adding substrates. Peixoto et al. [228] found that, based on 16 S rRNA gene analysis of bulk soils under conventional tillage (CT) and no-tillage (NT), the microbial community in NT soils more closely resembled that of nearby forested soils than that of the conventionally tilled soil. They attributed this to changes in soil structure, such as an increase in the mean weight diameter of aggregates under NT. Recently, Muruganandam et al. [206] found that while the composition of the microbial community varied between aggregates from tilled and NT soils, it did not differ among aggregate-size classes. Adopting these strategies promotes both sustainable agricultural systems and improved environmental health.

4.2.3 Maximize the presence of living roots

Incorporating practices such as perennial crops, relay cropping, diversified crop rotations, cover crops, and reduced fallow periods is vital for boosting living root presence in soil. Living roots provide crucial food for microorganisms, reducing erosion and enhancing nutrient cycling for optimal plant growth [289]. Lemongrass cultivation in Central Gujarat demonstrates how longstanding perennial plants enhance soil fertility with their dense root systems and help prevent erosion through their soil-binding capacity [126]. In a horti-silviculture system (dragon fruit + Malabar neem), fine root biomass, fine root production, and fine root turnover rate ranged from g m^{-2}, 109 to 160

g m^{-2} yr^{-1}, and 1.28 to 1.37 yr^{-1}, respectively [125]. The opportunistic and exploitative nature of fine roots enables them to quickly adjust to changes in water and nutrient supply [231]. The most effective results were achieved with the plantation of a two m-wide vegetative barrier of Guinea sp., which reduced runoff by 30%, soil loss by 66%, nutrient loss by 69%, and SOC loss by 65%, owing to its strong soil-binding capacity [241]. Relay cropping optimizes nutrient use and prevents soil degradation, especially evident in maize-legume systems (Punyalve et al., 2015; [277]). Diverse cropping and cover crops increase root density, microbial activity, and nutrient availability, and deter pests such as white grubs [240]. Reducing fallow periods or using fallow lands improves root density, soil organic matter, and water retention, supporting soil health and sustainable agriculture [33]. These practices collectively foster soil health, nutrient cycling, and sustainable agricultural productivity.

4.2.4 Reduced tillage

Implementing controlled traffic and minimum soil disturbance practices, such as zero tillage or minimum tillage, significantly improves soil health. Farmers frequently see tillage as a management technique for physically modifying soil to improve crop establishment. Numerous ecosystem services and soil processes are affected when soil physical characteristics are altered [37, 160, 163]. In the long term, excessive tillage techniques may negatively affect soil health. These methods regulate soil organic matter oxidation, enhance porosity, and deter weed seed germination [119, 120]. Combining zero tillage with residue retention boosts rainwater infiltration by 8.3 to 21.5% [247], reduces runoff erosive potential by 19.3% [158], enhances aggregate stability by 5.8% [211], and organic carbon content by 15.9% [117] Adhering to conservation agriculture principles, including soil cover and diversified cropping, reduces compaction, moderates soil temperature, and mitigates erosion [244]. Reducing synthetic pesticide use benefits beneficial organisms and pollinators; promoting organic or bio-pesticides improves soil biological activity [18]. Conservation agriculture practices have been shown to improve soil moisture content significantly, mean weight diameter (MWD), and water-stable aggregates (WSA > 0.25 mm), especially in the topsoil (0–15 cm). These practices also lead to notable reductions in bulk density and penetration resistance [118]. The extent to which low tillage and other practices affect the soil's physical characteristics can vary by soil type and climate. Parihar et al. [225] observed that in semi-arid sandy loam soils, water-stable aggregates increased by 16.1–32.5%, while bulk density and penetration resistance reduced significantly. Similar outcomes in sandy loam soil were also reported by Singh et al. [263] following five years of testing. In contrast, Somasundaram et al. [270] found that, after 4 years of testing, clay soil in central India's hot sub-humid climate showed a 10% increase in WSA and a 20% increase in MWD compared with conventional tillage. These practices collectively enhance soil porosity, preserve organic matter, reduce erosion, and foster sustainable agriculture while supporting soil health.

5 Regenerative agriculture approaches

RA is a set of farming practices and concepts that repair watersheds, boost ecosystem services, enrich soils, and increase biodiversity, according to Terra Genesis International (Fig. 3). The goal of RA is to reverse the existing worldwide patterns of atmospheric accumulation by capturing carbon in soil and aboveground biomass. RA may be considered

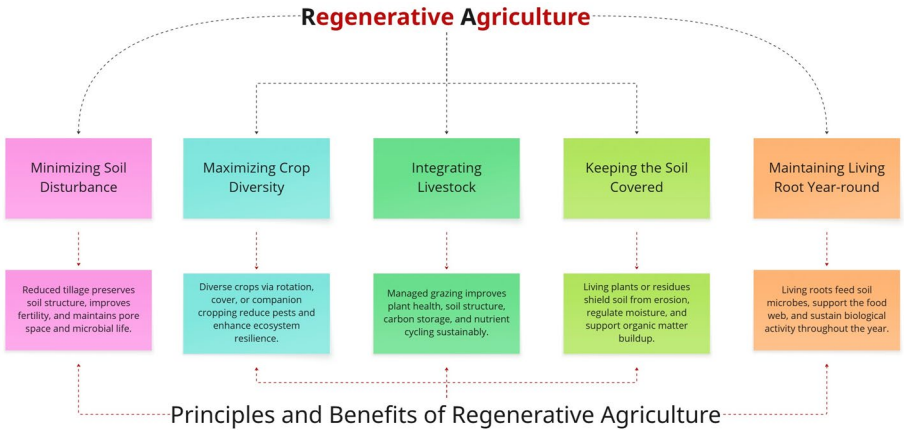


Fig. 3 Regenerative agriculture approaches

as an important tool for sustainable agriculture along with other approaches like climate smart agriculture (CSA), conservation agriculture (CA), organic farming etc. Tittonell et al. [283]. The goal of RA is to produce more from less: "less land area, less input of chemicals, less use of water, less emission of greenhouse gases, less risk of soil degradation, and less use of energy-based inputs" [162].

It brings together economic, social, and environmental dimensions by emphasizing three core objectives: (1) achieving sustainable growth in agricultural productivity and incomes, (2) strengthening adaptation and resilience to climate change, and (3) reducing or eliminating greenhouse gas emissions [78]. Soil conservation and management are vital for food security, but they face challenges due to climate change, including changes in temperature and precipitation [42, 260]. Conservation agriculture is key to establishing sustainable agro-ecosystems. Practices such as no-till systems, mulch crops, cover crops, nutrient management, and crop rotation contribute to soil and water conservation, carbon sequestration, and adaptation to local climate conditions [161]. A recent study found that a 1% increase in SOM enhanced available water holding capacity (AWHC) by up to 1.5%, depending on soil texture and mineralogy, with the effect more pronounced in sandy soils [169]. Understanding soil biodiversity, including bacteria, fungi, and other organisms, is crucial for effective soil management and sustainable agriculture [115].

Through carbon sequestration and enhanced nitrogen usage efficiency, integrated nutrient management (INM) promotes soil fertility and agricultural productivity while mitigating the effects of climate change [219]. Across the cropping system under study, the increase in crop yields under INM over traditional nutrient management ranged from 1.3% to 66.5%. In contrast to conventional nutrient management in the rice-wheat cropping system, following straw centralized returning reduced CH_4 emissions in the rice season by 27.5% in China [168]. Practices such as mulching improve soil moisture, structure, and temperature, and reduce soil and nutrient loss [219]. In addition to increasing soil fertility, cover crops can sequester carbon. Their widespread use might lower agricultural GHG emissions by 10%, which is equivalent to the use of no-till or other cropping techniques [145]. Reducing agricultural greenhouse gas emissions is achievable through improved practices in crop establishment, nutrient management, irrigation, tillage, and residue management [11, 195]. Water conservation methods, drought-resistant varieties, optimized irrigation, and fertilizer management

are recommended for climate-resilient agriculture [216]. According to a meta-analysis of 740 paired measurements from 90 peer-reviewed articles, NT increased barley yield by 49%, reduced the global warming potential at acidic soil sites, and demonstrated potential to lower greenhouse gas (GHG) emissions in dry climates compared with conventional tillage [107]. In just five years, damaged cropland was repaired, and carbon stocks in tropical sandy soils were increased through crop rotation and mixed-species cover crops without tillage [57]. Crop rotation dramatically boosted soil microbial biomass C and N by 20.7% and 26.1%, respectively, according to a meta-analysis of 122 studies on the impact of crop rotation on soil biological characteristics [186]. After 50 years of applying chemical fertilizers, Khan et al. [147] found that SOC declined even when crop residues were returned to the soil, because crops removed an additional 60% to 190% of the nitrogen supplied, resulting in a net depletion of soil organic matter and long-term soil fertility.

6 Scaling CSA-capital, policy, and principles for sustainability

Climate-smart agriculture aims to enhance food security by addressing climate change challenges and evolving agricultural practices [174]. Climate-Smart Agriculture operates through two complementary pathways to ensure food security under climate change (Fig. 4; [64]). Production improvement focuses on developing resilient crop varieties, enhancing soil quality, and strengthening irrigation to support adaptation to warming.

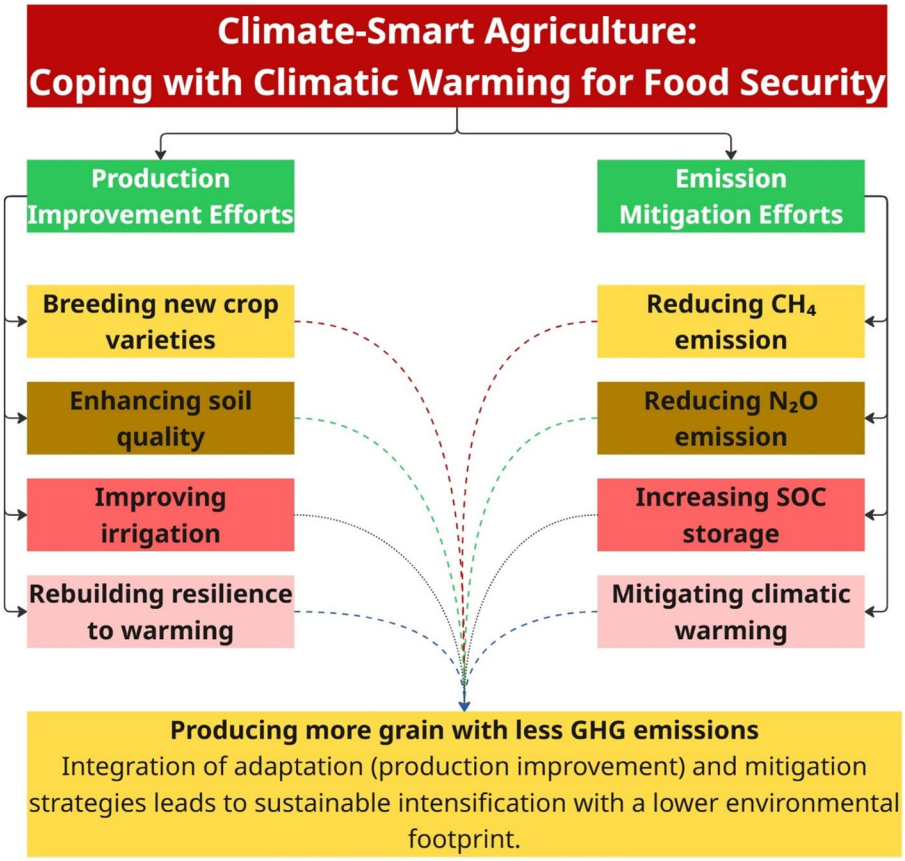


Fig. 4 Framework of climate smart agriculture associated with cropping system innovations in China. (Source: [64])

Emission mitigation emphasizes reducing CH_4 and N_2O emissions and increasing soil organic carbon to curb climate warming. Together, these strategies aim to produce more grain with lower greenhouse gas emissions. Countries need to align their adaptation strategies with the Paris Agreement and enhance resilience by implementing strong, well-integrated agricultural policies [293]. Climate models project economic losses up to US$1.01 trillion by 2050, primarily affecting consumers through higher food prices [222]. Farmers often lack region-specific information on climate impacts and adaptation options, limiting their ability to cope with shifting growing seasons and pest pressures. Knowledge exchange and access to climate data, forecasts, and early warning systems are essential for effective response strategies [251]. However, only 25% of survey respondents reported using resources from the Global Alliance for Climate-Smart Agriculture (GACSA), indicating limited dissemination even among actively involved stakeholders.

Government policies play a critical role in promoting CSA by incentivizing sustainable practices, supporting resilient crop research, improving water and soil management, and encouraging agroforestry [272]. Despite this, GACSA members cited limited funding and weak extension services as major barriers. Strengthening GACSA's collaboration with Regional Alliances and enhancing capacity-building efforts can bridge gaps between policy, knowledge, and practice. Increased participation in CSA forums and workshops can further improve adoption. While CSA supports adaptation and resilience, it does not address the root causes of climate change. Most smallholder farmers in developing countries contribute minimally to global GHG emissions, yet they remain highly vulnerable. Meaningful climate action demands coordinated global efforts across all sectors to achieve substantial emission reductions.

6.1 Enabling policy frameworks to advance RA and CSA

Policymakers play a crucial role in advancing RA by offering incentives and enacting supportive regulations at both national and international levels. Aligning regenerative practices with global policy goals, such as reducing agriculture's negative effects on biodiversity, ecosystems, and human health, can further accelerate adoption. International initiatives like France's "4 per 1000" campaign aim to increase soil carbon by 0.4% annually over a decade [1], while regenerative approaches are recognized in international frameworks and sustainability assessments [79]. The FAO's 2019 report highlights the ecological benefits of agroforestry. In response to mounting evidence of biodiversity loss and ecosystem decline, the UN declared 2021–2030 the Decade on Ecosystem Restoration [109, 110, 223]. Wolff et al. [302] proposed safeguarding 28% of the Earth's most biodiverse and carbon-rich lands, up from 15% in 2018 [80]. Further, Springmann et al. [271] used the planetary boundary framework to examine sustainable land use, while the World Bank [303] evaluated strategies to improve soil carbon sequestration.

At the national level, governments can strengthen RA adoption by introducing policies that financially reward ecosystem services and encourage practices that reduce emissions. The UK's 25-year environmental plan emphasizes net biodiversity gains, reduced pesticide use, and payments to farmers for restoring natural capital, reinforcing the "polluter pays" principle. India's Paramparagat Krishi Vikas Yojana (PKVY), launched in 2015, supports organic farming. Regulatory tools can also help, such as banning wetland removal or deforestation, mandating biodiversity practices, and restricting harmful

pesticides like neonicotinoids [101, 149]. India's National Adaptation Fund for Climate Change (NAFCC), launched in 2015, aids vulnerable states and union territories in climate adaptation efforts aligned with the State and National Action Plans on Climate Change (SAPCC and NAPCC). National Bank for Agriculture and Rural Development (NABARD) serves as the National Implementing Entity, facilitating project development, approval, funding, monitoring, and capacity-building among stakeholders [281].

6.2 Institutional capacity and coordination mechanisms for expanding CSA

Improving productivity and profitability in farmers' practices is essential for climate-friendly agriculture, which depends on efficient labor and suitable crop choices. It also involves implementing reforms to create and strengthen legal and specialized institutions that empower local communities, promote diverse livelihoods, and support problem-solving strategies. Supportive organizations and policies should be available to help manufacturers adopt new practices. These efforts should involve all relevant stakeholders, including producers, agribusinesses, traders, buyers, and public authorities, to ensure collective participation and cooperation in reaching climate-smart agriculture goals [49].

From farmers at the community level to policymaking bodies, the role of institutions in enhancing Climate Smart Agriculture in India has been growing. These institutions are vital for producing and disseminating relevant information and for assisting the farming community in putting it into practice. Hence, strengthening institutions, including financial, land, legal, community, insurance, and information organizations, is crucial for supporting farmers in adopting climate-smart agriculture practices (Table 2). At the local, national, and international levels, agricultural, climate change, social protection, and food security groups must communicate effectively with one another [262]. To improve cooperation and information exchange, these organizations can host workshops, examine policies, and lead conversations (NAAS, [15, 177, 210, 276]). Efforts should focus on building the national governments' capacity for international climate change conferences and fostering cooperation with local authorities. Strengthening institutions and communication channels enables stakeholders to develop coordinated strategies, leading to the successful implementation of climate-smart agriculture and resilient agricultural systems. A strong institutional structure can effectively manage and balance such technological transfers because farmers lack the resources to invest in and adjust to these new approaches for the success of CSA, as well as to access markets and loans [184]. Table 2 highlights several organizations involved in designing and implementing CSA-related programs and facilitating technology transfer, all of which play key roles in expanding and disseminating India's CSA expertise.

6.3 Financial instruments and support systems for CSA

Incorporating innovative financial systems that blend public and private funding is critical for the effective implementation of climate-smart agriculture (CSA). Mechanisms such as the Green Climate Fund and tools such as Nationally Appropriate Mitigation Actions (NAMAs) and National Poverty Reduction Action Plans (NPRAPs) help align national programs with diverse funding streams, including public budgets and development assistance. Climate change considerations must be embedded into budget planning and organizational strategies to enable effective mitigation. For agricultural

Table 2 Leading institutions (Government and Semi-Governmental) and centers of excellence in Climate Smart Agriculture (CSA) with their mandates and logo

S. No.	Name of institution	Vision	Website link
A. Public Institutions (Government and Semi-Governmental)			
1.	National Initiative in Climate Resilient Agriculture (NICRA), New Delhi	To showcase site-specific technology packages on farmers' fields for adapting to present climate hazards and to increase the resilience of Indian agriculture to climatic variability and climate change through the development and use of improved production and risk management technologies.	https://www.india.gov.in/website-national-innovations-climate-resilient-agriculture
2.	National Institute of Rural Development and Panchayati Raj (NIRD & PR), Hyderabad	The goals of NIRD&PR are to improve the effectiveness of rural development staff, encourage technology transfer through its Technology Park and social laboratories, raise awareness of environmental issues, and concentrate on policies and programs that help the impoverished in rural areas. In addition to serving as a knowledge base on rural development, NIRD serves as a "think-tank" for the Ministry of Rural Development, helping the Ministry formulate policies and select solutions for rural development that would bring about change.	https://nirdpr.org.in/
3.	National Institute of Agricultural Extension Management (MANAGE), Hyderabad	The institute is tasked with working in the following directions under the MANAGE mandate: 1. Establishing connections between important national, regional, state, and international organizations involved in agricultural extension management 2. Acquiring knowledge of agricultural extension management policies and procedures 3. Establishing cooperative relationships with domestic and foreign organizations to exchange academic resources 4. Creating and encouraging the use of contemporary management techniques to increase agricultural extension organizations' efficacy 5. Setting up need-based training for middle- and senior-level agricultural extension employees 6. Performing research on agricultural extension management that is problem-oriented 7. Acting as a global documentation hub for gathering, preserving, processing, and sharing data on topics about agriculture/of agriculture	https://www.manage.gov.in/
4.	National Academy of Agricultural Research Management (NAARM), Hyderabad	The Academy's primary responsibilities include advocating for policies for the National Agricultural Research and Education System (NARES) and strengthening agricultural research, education, and extension education institutions.	https://naarm.org.in/home/
5.	National Gender Resource Centre in Agriculture (NGRCA), New Delhi	To internalize gender-specific interventions and make sure that agricultural policies and programs are fully engendered and reflect the national commitment to women's empowerment, the National Gender Resource Centre in Agriculture (NGRCA) serves as a focal point for all gender-related activities and issues in agriculture and related sectors, both inside and outside the Department of Agriculture & Farmers Welfare. It also implements and supports training, research, and advocacy to mainstream gender issues in agriculture and natural resource management.	https://www.krishivistar.gov.in/Ngrca.aspx

Table 2 (continued)

S. No.	Name of institution	Vision	Website link
6.	Indian Institute of Millets Research (IIMR), Hyderabad	To use value addition to turn millets from subsistence farming into a globally competitive, climate-resilient, nutri-cereal enterprise that can meet the nation's needs for food, feed, fodder, nutrition, and biofuel in order to promote fair prosperity.	https://www.millets.res.in/
7.	Institute of Rural Management Anand (IRMA), Gujarat	To support rural residents' socioeconomic growth in a sustainable, environmentally responsible, and equitable manner by means of professional management, leadership in the creation of cutting-edge knowledge, educational innovations, the pursuit of the sustainability ethical imperative, and policy-oriented research.	https://www.irma.ac.in/
B. Private Institutions (NGOs, for-profit companies, incubation centers etc.)			
1.	The Energy and Resources Institute (TERI), New Delhi	With creative approaches to waste reduction and reuse, as well as the conservation and efficient use of energy and other resources, TERI aims to accelerate the shift to a cleaner and more sustainable future.	https://www.teriin.org/
2.	Crop In, Bengaluru	Cropin's ability to develop and provide multifaceted predictive intelligence has the potential to revolutionize the entire crop ecosystem.	https://www.cropin.com/
3.	Digital Green, New Delhi	Through the development of complementary solutions and the improvement of our digital extension approach, Digital Green aims to raise the income of smallholder farmers and enable them to escape poverty by utilizing technology and forming partnerships at the grassroots level.	https://digitalgreen.org/ourwork/india/
4.	Centre for Sustainable Agriculture, Secunderabad	As a national organization, CSA aims to guide India's agricultural sector toward ecological and economic sustainability. By researching agro-ecological farming practices and their effects, helping farmers and consumers make the switch to organic, creating thriving rural businesses and producer organizations, improving the last mile delivery of support services, and strengthening the governance of public policy systems, it aims to care for those who feed the country.	https://csa-india.org/
5.	M. S. Swaminathan Research Foundation (MSSRF), Chennai	Its goal is to apply contemporary research more quickly for rural and agricultural development that is sustainable. MSSRF takes a pro-poor, pro-women, and pro-nature stance and concentrates on rural and tribal areas.	https://www.mssrf.org/
6.	BAIF Development Research Foundation, Pune	The goal of BAIF is to give rural families—especially those from underprivileged backgrounds—the chance to engage in profitable self-employment, guaranteeing a stable income, a better environment, an enhanced standard of living, and moral principles.	https://baif.org.in/
7.	Professor Jayashankar Telangana State Agricultural University (AgHub, PJTSAU), Hyderabad	• Training and human resources required for home science, agriculture, agricultural engineering and technology, and related fields. • Perform fundamental and applied research to create better cultivars and technologies for the advancement of sustainable agriculture. • Encourage on-farm research and technology evaluation, improvement, and knowledge sharing via collaborative methods. • Create alliances and connections with agro-industries, the rural development sector, and national and international educational, research, and development institutions.	https://pjtau.edu.in/

investments to attract climate finance, interventions must be quantifiable, reportable, and verifiable. Evidence-based data on CSA is essential to bridge global agreements with national actions.

The 2030 Agenda, including the Sustainable Development Goals (SDGs) and the Paris Agreement, highlights agriculture's role in ensuring food security and building climate resilience. Instruments like NAPA (National Adaptation Programme of Action), PNA (National Adaptation Plan), and NAMA facilitate the integration of climate and development goals. The Sendai Framework for Disaster Risk Reduction also underlines the increasing disaster risks due to climate change. Achieving CSA transitions requires political commitment, cross-sectoral coordination, and inclusive, participatory approaches. Policies and legal frameworks must support indigenous knowledge, gender equity, and equitable access to resources.

Public-private partnerships play a pivotal role, especially for smallholder farmers, enabling cross-border collaboration and innovation in CSA practices. In India, the government has introduced several initiatives to support CSA across the full agricultural spectrum. These include central sector and centrally sponsored schemes like the Soil Health Card (SHC), Per Drop More Crop (PDMC), Crop Residue Management (CRM), Micro Irrigation Fund (MIF), Sub-Mission on Agricultural Mechanization (SMAM), National Beekeeping and Honey Mission (NBHM), Namo Drone Didi, and Digital Agriculture. These programs address infrastructure, technology, mechanization, irrigation, inputs, credit, insurance, marketing, and agribusiness support. The financial pathways and institutional linkages that channel resources through these schemes to promote CSA adoption are illustrated in Figs. 5, 6 and 7.

7 Dimensions of soil security framework

The soil security framework's dimensions are essential for understanding and addressing a range of land management and sustainability-related issues. Capability, condition, capital, connectedness, and codification are among these dimensions (Fig. 8). As with other notions such as food and water security, a set of dimensions must be identified and developed to frame this idea. These dimensions should take into consideration the quantity, quality, and accessibility of the soil. It's also critical to understand that this idea is not being developed in a vacuum; comparable ideas of soil protection, health, and quality have also been put forth to address the necessity of managing and preserving the state of the soil [41, 143]. Understanding soil security threats is essential for developing

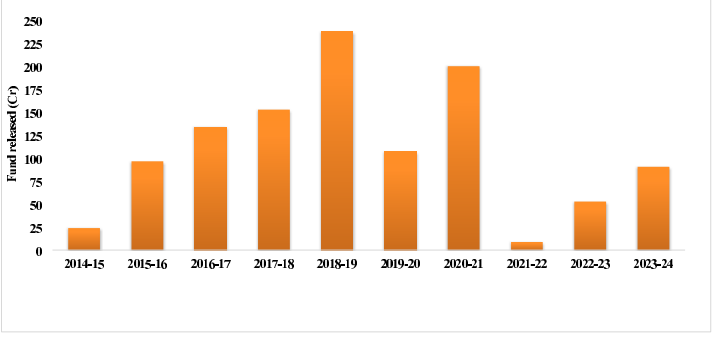


Fig. 5 Funds released under Soil Health Card Scheme (SHC) to promote Climate Smart Agriculture [191]

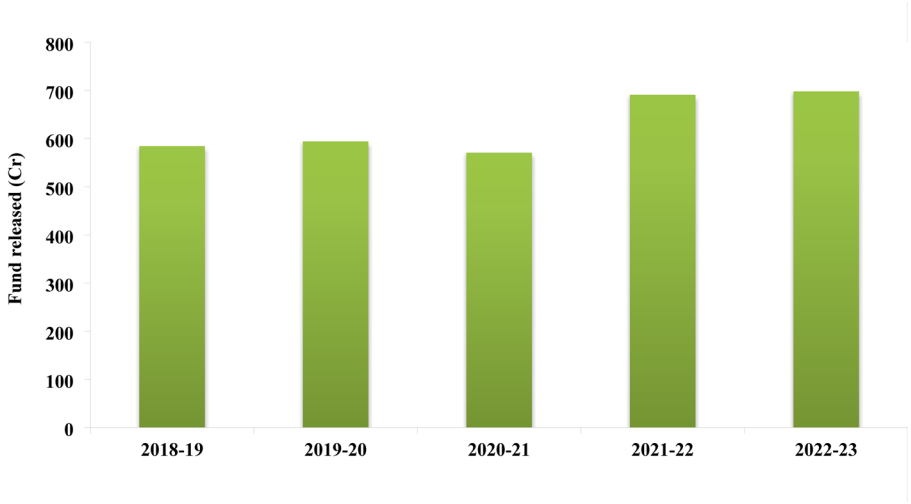


Fig. 6 Funds released under Crop Residue Management Scheme (CRM) to promote Climate Smart Agriculture [191]

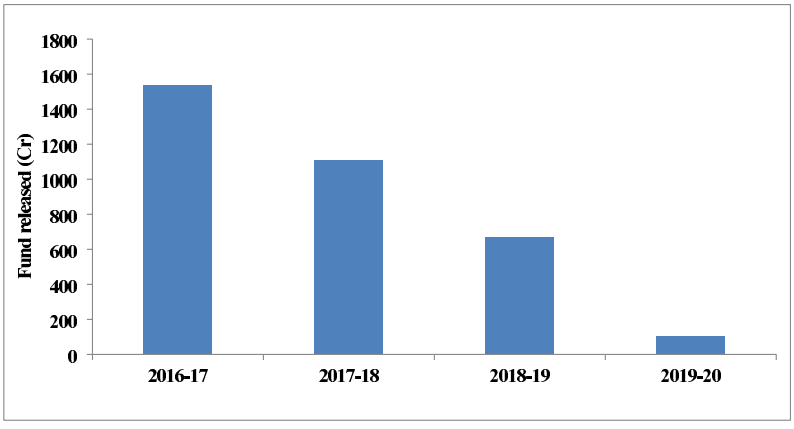


Fig. 7 Funds released under National Beekeeping and Honey Mission (NBHM) to promote Climate Smart Agriculture [191]

effective soil protection strategies. Figure 8 illustrates the major soil security threats across various dimensions, as outlined by McBratney et al. [185].

8 Cropping system and soil health

Amsili et al. [12] reported that while cropping systems significantly influenced soil organic matter (SOM), soil texture accounted for a larger share of SOM variance (21.8%) compared to cropping systems (8.2%). Table 3 illustrates how different agricultural strategies affect soil health. Crop rotation improved soil microbial biomass carbon (MBC) and nitrogen (MBN) by 13.43% and 15.84%, respectively, over continuous monoculture. Fungal biomass increased significantly (45.50%) when diverse crops were introduced into bean monoculture systems, particularly in regions with mean annual precipitation between 600 and 1000 mm (15.72%). Rotation also raised soil bacterial diversity, as measured by Shannon's index, by 7.68% compared to monoculture [176]. Burle et al. [45] found that legume-based systems enhanced effective cation exchange capacity (ECEC) and soil organic carbon, especially in the top 2.5 cm of soil. However, clover-based

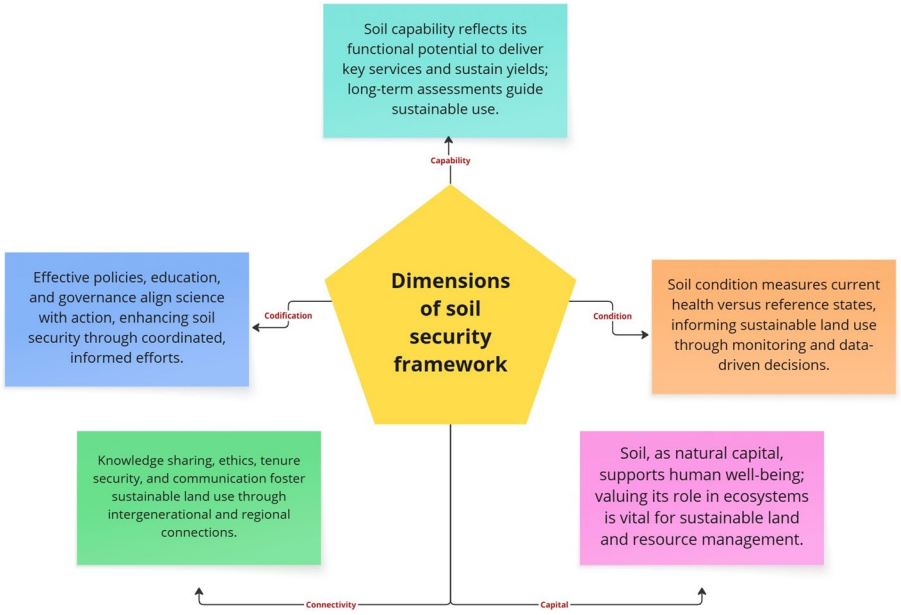


Fig. 8 Dimensions of soil security framework

systems also led to greater soil acidity at depths of 2.5–7.5 cm. In particular, the clover + Spergula/maize rotation decreased soil pH by 0.1 units per year in the 2.5–7.5 cm layer, thereby influencing ECEC.

Ahirwal et al. [7] highlighted land-use change impacts, with soil organic carbon (SOC) declining by 84% in AMS and 50% in AGS, while soil CO_2 efflux rose by 35% in AGS but dropped by 43% in AMS compared to native forest soils (NFS), reflecting site-specific microbial and plant interactions. Integrated soil management—encompassing fertilization, organic amendments, improved germplasm, and local adaptation—enhances crop productivity and nutrient use efficiency. Conservation agriculture practices such as residue retention, cover cropping, no-till systems, and optimized nutrient management contribute to carbon sequestration, water conservation, climate adaptation, and sustainable intensification. Soil biodiversity, comprising archaea, bacteria, fungi, protozoa, nematodes, and macrofauna, is essential for ecosystem functioning. Effective soil management hinges on understanding the dynamics of soil biodiversity, its influence on soil properties, and its role across climates and land-use systems. Tesfaye, [279] emphasized integrated nutrient management to sustain soil fertility through residual nutrient effects. Overall, boosting soil organic matter, conserving soil biodiversity, and adopting holistic management strategies are pivotal to climate-resilient, sustainable agriculture.

8.1 Crop diversification

Agriculture is facing serious challenges from climate change, particularly for subsistence farmers in areas with delicate ecosystems. To enhance agricultural resilience, adopting diverse production techniques is crucial. In climate-vulnerable systems, practices such as agroforestry, crop diversification, livestock integration, and planned fallow periods can enhance resilience, functional diversity, and overall agricultural stability (Table 4). It has been demonstrated that crop diversification reduces the impact of microclimatic shifts on crop production. While limited research specifically links crop diversity to

Table 3 Effect of different cropping systems on soil health

Sl. No.	Cropping system	Region	Effect on soil health	References
1.	Wheat-soybean under INM	Central India	Increased phosphate availability, organic matter, and non-symbiotic bacterial activity.	Behera et al. [29]
2.	Lupin-strawberry	Spain	Enhance Fe^{2+} particularly oxalate-extractable Fe (Feo).	Velazquez et al. [292]
3.	Sorghum + pigeon pea	Uttar Pradesh, India	HCN, which is released by sorghum, helps control soil-borne diseases (*Fusarium udum*) by up to 70%. It also improves microbial activity, decreases soil moisture loss, and increases N availability during the early stages of sorghum growth.	Ali, [10] Natarajan et al. [213]
4.	Soybean-wheat under INM	Central India	Organic manure increases the availability of soil nutrients including N, P, and K, promotes the growth of soil microbes, and improves soil structure.	Pedaprolu et al. [227]
5.	Khejri-based agro-forestry system	Rajasthan, India	The soil microbial population becomes more active, the deep taproot system aids in holing soil particles, increased dehydrogenase and alkaline phosphatase activity, and protection against wind and splash water erosion.	Yadav et al. [309]
6.	Maize + black gram	West Bengal, India	Increased more C and N, conserved more soil water, microbial activity and land use efficiency (42%).	Sarkar et al. [253]
7.	Rice-cabbage-green gram	West Bengal, India	The cultivation of rice provides more soil moisture for cabbage, improvement in quality of rice grain (amylose content 25.27%, hulling 77.67%, milling 66.67%) and green gram enhanced N, improve soil porosity and SOC.	Acharya and Mondal, [4]
8.	Groundnut-finger millet	Karnataka, India	SOC, CEC, and EC all increased. More availability of N was seen than of P and K. Planting groundnuts first reduces the amount of nitrogen lost compared to finger millet and keeps P and N in balance.	Srinivasarao et al. [273]
9.	Bhabbar grass + black gram	India	Bhabbar grass stops soil erosion by holding soil particles in place with its roots and increased N content for healthy canopy growth. Intercropping reduces production by as much as 68.7% when compared to black gram grown alone.	Mahapatra, [178]
10.	Dhaincha as green manure in rice field	West Bengal, India	The soil's microbial population has improved, the bio-organic amendment raised the amount of nitrogen available in the rice field when compared to fertilizers alone, and the positive influence of zinc and the microbial population grew by up to 25%.	Roy et al. [249]
11.	Wheat-green gram	Delhi, India	Long-term supply of mineral N in soil; higher levels of Olsen P in wheat during blooming and maturity; increased soil K content from flowering to pod formation stage; and 40–46% increases in dehydrogenase and phosphatase activity.	Moharana et al. [197]
12.	Groundnut-toria + gobhi sarson	Punjab, India	Fix nitrogen and solubilize phosphorus, and reduce the number of root-knot nematodes in the soil by up to 50%.	Sinha et al. [267]
13.	Aman rice-chickpea	Bangladesh	Soil moisture utilization, increase N content in soil and best rhizosphere activity with increased yield of 47%.	Musa et al. [207]
14.	Groundnut-rice-black gram	Tamil Nadu, India	Increased SOC, accessible N, P, K, and S content, as well as improved soil moisture utilization.	Porpavai et al. [235]
15.	Rice-rice-green gram	Tamil Nadu, India	Enhancement of the SOC content and soil structure.	Porpavai et al. [235]
16	Pearl millet-wheat	North-Western India	Applying organic fertilizers and manures to the pearl millet-wheat cropping system for a continuous 25 years raised the levels of SOC, MBC, and enzymatic activities.	Kumari et al. [156]

Table 4 Diversification of agricultural systems and potential benefits for farmers under climate change. Source: Modified from Lin [172]

Diversification types	Description of diversification	Main benefits	Examples
Improved structural diversity	Strip intercropping, which involves growing many crops in strips that are both wide enough to allow for autonomous cultivation and narrow enough to interaction, is one example of how it increases the structural diversity of crops grown in the field.	Pest suppression	During harvest, natural enemies can spread from harvested strips to nearby non-harvested ones when alfalfa is cut in strips [105].
Genetic diversity in monoculture	Cultivating different cultivars of a species in a mono-cropping.	Disease suppression Increased production stability	Fungal blast occurrence is decreased by rice cultivars' genetic diversity (Zhu et al., 2000). Mean income and income stability were positively correlated with increased genetic diversity [68].
Crop rotations	Temporal diversity through crop rotations.	Disease suppression Increased production	Disease cycles are disrupted by switching between broadleaf and cereal crops and by altering stand densities [150]. Increased crop rotations of nitrogen-fixing and cover crops improved the primary crop's productivity by manipulating diversity [269].
Polyculture	Cultivating a field containing two or more crop species and wild variants. Agricultural diversity in terms of time and space.	Disease suppression Climate change buffering Increased production	Multiple-species grassland areas are planted to reduce the spread of disease [194]. Climate stress allowed more biologically complex systems to flourish, including wild varieties and the temporal and spatial variety of crops [278]. More constant feed and fodder production was seen in grassland plots with a higher in-field species diversity [282]. Grassland plots having more species diversity in the field produced higher and more stable biomass yields over time, owing to complementary resource use and reduced yield fluctuations across seasons [234].
Diversify crop field with non-crop vegetation	Crops are grown in and next to vegetation banks or weed strips.	Pest suppression	For natural enemies, grassland or refugia established at field edges (beetle banks) served as an overwintering refuge. The suppression of pests by applying black and white mustard to the edges of sweet corn fields deterred pests from entering the field [243].
Alternate land use system (Agroforestry system)	Growing crops and trees together.	Pest suppression Climate change buffering	There are fewer leaf beetle pest outbreaks among willow trees cultivated in their native settings [61]. Birds were more abundant as natural enemies to suppress agricultural plant larvae when there was greater diversity in the shade [230]. In shade environments, higher ant diversity and abundance led to better control of coffee berry borer [14]. Increased crop buffering against changes in temperature and precipitation was a result of more shadow cover [171]. More shade tree cover resulted in less storm damage and more protection from storm events [233].

Table 4 (continued)

Diversification types	Description of diversification	Main benefits	Examples
Mixed landscapes	Creation of more diverse, multi-ecosystem landscapes on a bigger scale.	Pest suppression Increased production	In complex landscapes with hedgerows and woodland patches scattered throughout fields, larval parasitism rates were greater. Higher rates of rapeseed pollen beetle parasitism were seen in oilseed rape crops that were next to vast, complex, and structurally rich ancient fallows. The best diversity and profitability solutions were achieved by a combination of crop rotations, intense managed grazing, and organic cropland (Boody et al., 2009).

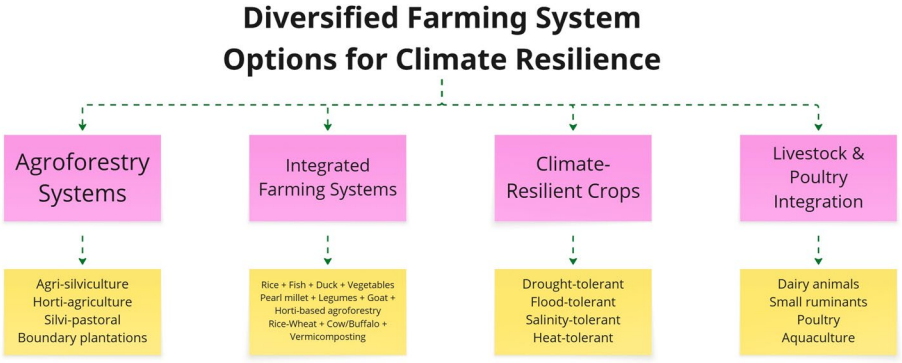


Fig. 9 Crop diversification options

reduced vulnerability, studies on agroforestry, polycultures, and livestock diversification demonstrate resilience across various contexts (Fig. 9). For example, agroforestry systems in India provide economic security for disadvantaged farmers. Crop yields are vulnerable to fluctuations in temperature and precipitation during critical stages such as flowering and fruit development. Diverse farming systems that incorporate improved seeds, soil conservation, and crop diversity can increase farmers' incomes and soil health. Ecologically complex systems, wild crop varieties, and diverse cropping systems can mitigate local climate variability, reduce erosion, and enhance production resilience.

Studies in Tanzania, Sweden, Nicaragua, and Southern Africa highlight the benefits of diversifying agricultural systems in mitigating climate variability and improving food security. Crop diversification balances soil biodiversity, enhances nutrient utilization efficiency, reduces pathogens, and increases crop yields and associated biodiversity. However, challenges such as market access, irrigation availability, land size, land rights, energy poverty, policy support, and economic incentives hinder the adoption of crop diversification. Biotech's focus on drought-resistant crops and misconceptions about monoculture biomass yield also impede the transition to diverse systems. In general, improving soil health and ecosystem services, ensuring food security, and enhancing climate change resilience all depend on promoting a range of agricultural strategies. Eliminating barriers while encouraging legislative support will facilitate the widespread adoption of diverse farming systems. Furthermore, the contribution of crop diversity to attaining the SDGs is emphasized in Table 5.

Table 5 Contribution of crop diversification to the achievement of the Sustainable Development Goals (SDGs)

Sustainable Development Goals	Types of advantages	Specific advantages of crop diversification	References
Goal 15: (Life on land)	Environmental resilience	• Prevents soil exploitation • Lessens soil erosion • Offers habitat • Maintaining a diversified vegetation cover that offers environmental services • Preventing the need for forest removal and the resulting environmental impacts • Improving nutrient uptake and cycling that supports productivity on current plots	Ellis, [76] Joshi et al. [139] Perz, [232] Lin, [172]
Goal 13: (Climate action)	Climate change mitigation and adaptation	• Lessens the need for nitrogen fertilizer, which lowers greenhouse gas emissions • Makes the cropping system more resilient to waterlogging and drought • Enhances the stability of the yield • Because climate events impact different crops in different ways, it can act as insurance against rainfall variability • Water stress intensifies the relative productivity disparities between monocultures and polycultures • By increasing the ability to control pest outbreaks and minimize the spread of pathogens, which could get worse in future climate scenarios, and by protecting crop production from the consequences of increased climate variability and extreme events, crop variety can increase resilience	Joshi et al. [139] Nhemachena and Hassan, [215] Truscott et al. [288] Lin, [172] Antwi-Agyei et al. [13] Gajigo, [83] Mijatovic et al. [190] Alemayehu and Bewket, [9]
Goal 12: (Sustainable production)	Crop productivity	• Boosts soil fertility • Boosts production and stability of production • Can boost yields generally and the main crop's productivity • Lowers the risk brought on by seasonal variables	Adger et al. [5] Orindi and Eriksen, [221] Lin, [172] Mugendi Njeru, [203] Makate et al. [179]
	Minimize pest and diseases	• Suppresses pest outbreaks and slows the spread of pathogens, which can increase resilience in a variety of ways • Manages diseases and pests • Gives beneficial insects a place to live, which can help reduce pests by making host crops less visible for parasites to colonize	Truscott et al. [288] Lin, [172] Shoffner and Tooker, [258]
	Safety against market oscillations	• Financial risk is decreased • price risk and output swings are mitigated • Crop diversification is a risk avoidance tactic for abrupt changes in crop yield or price	Ellis, [76] Ashfaq et al. [16] Gajigo, [83] Coromaldi et al. [59] Yu, [312]
Goal 8: (Decent work and economic growth) Goal 10: (Reduced inequalities)	Increase income and employment	• Increases the range of agricultural operations • Stabilizes farming income • Boosts economic stability • Contributes to greater economic stability, stable farming revenue, and a variety of agricultural practices • Stable jobs through a longer on-farm season • Produces job openings • Agriculture-related income is greatly increased by crop diversity	Guvele, [94] Joshi et al. [139] Huang et al. [106] Birthal et al. [36] Makate et al. [179]

Table 5 (continued)

Sustainable Development Goals	Types of advantages	Specific advantages of crop diversification	References
Goal 5: (Gender equality)	Gender equality	• Diversification in agriculture gives women producers more power	Joshi et al. [139]
Goal 2: (Zero hunger) Goal 12: (Sustainable consumption)	Nutrition and food security	• The production of fruits and vegetables improves food security and, eventually, people's anemia status (especially for women of childbearing age) • Crop diversification also directly affects food availability and nutrition • More varied production systems may result in more varied household diets	Lin, [172] Jones et al. [133] Kabunga et al. [140] Makate et al. [179]

8.2 Crop rotations

Long-term dependence on monoculture systems negatively affects soil microbial communities by reducing the diversity of beneficial microorganisms and promoting the buildup of specific pathogens. Predominant fungal phyla include *Ascomycota* (62.5%), *Glomeromycota* (11.1%), *Mucoromycota* (8.9%), and *Basidiomycota* (6.8%), while dominant bacterial phyla are *Pseudomonadota* (35.0%), Actinomycetota (21.1%), and *Bacillota* (10.1%) [22, 112, 299]. Monoculture disrupts ecological balance, reduces productivity and quality, and intensifies pest and disease outbreaks over time [124]. Climate change exacerbates these challenges by shifting species phenology and distributions, increasing pest pressure, and altering ecosystem dynamics [72].

In contrast, crop rotation enhances soil resilience and reduces erosion caused by extreme weather. For instance, in Jiangxi province, rotating alfalfa with corn increased yield by 1.5 t $acre^{-1}$ and improved soil aggregate stability by 35% [91]. Wang et al. [299] noted that rotation encourages macro-aggregate formation, improving erosion resistance. It also enhances soil structure, water dynamics, biological functions, and crop resistance to climate stress. Notably, rotation improves agroecosystem biodiversity by increasing community composition (61%), bacterial species richness (69%), and soil multi-functionality (52%) ([25]; Li et al., 2021). Rotation also increases water-use efficiency, reduces erosion, and improves water retention and drainage [6, 164, 298]. Its effectiveness is influenced by crop family, frequency (≥ 1 season), duration (up to 1 year), site history, and soil quality [31, 282]. Practices such as intercropping and crop rotation with suitable genotypes further enhance productivity and soil health [84, 311]. According to Xie et al., [305], intercropping Bletilla striata with peach trees in a silvo-medicinal system markedly increased soil nutrient availability, with total N, available N, available P, available K, and nitrate-N rising by 6.85%, 64.91%, 161.71%, 106.53%, and 16.82%, respectively, compared with monoculture. This system also elevated microbial activity, with notable increases in *Acidobacteriota* (0.17 ± 0.05 vs. 0.06 ± 0.03), *Chloroflexi* (0.12 ± 0.04 vs. 0.02 ± 0.00), and *Basidiomycota* (0.21 ± 0.12 vs. 0.09 ± 0.03). Similar findings in orchard intercropping systems confirm improvements in microbial richness [60, 70, 315]. In summary, crop rotation is a vital agroecological practice that enhances soil health, increases biodiversity, improves climate resilience, and boosts overall productivity, making it integral to sustainable farming systems (Table 6).

Table 6 Benefits provided by the crop rotation. Source: Yu et al., 2022

Particular	Benefit	Source
Water dynamics	• More efficient use of agricultural water and soil water storage • Decrease in surface water runoff	Trenberth, [286] Adiku et al. [6] Cebecauer and Hofierka, [47] Laloy and Bielders, [164] Baumhardt et al. [28] Rossato et al. [248] Cornelis et al. [58] Hatfield and Dold, [96] Han et al. [95] Wang et al. [298]
Soil health	• The soil's enhanced chemical and physical qualities • Boosts nutrient consumption, activates nutrients, and increases soil enzyme activity. • Increases the number of helpful bacteria while decreasing the number of pathogenic bacteria, improving the microecology of the soil.	Qiu et al. [239] Guo et al. [92] Troeh et al. [287] Perez et al. [229] Sarwar et al. [254] Jiang, [124] Meng et al. [188] Chen et al. [52] Marburger et al. [182] Wright et al. [304] Wan et al. [297] N'Dayegamiye et al. [208] Wang et al. [299] Xu et al. [307] Hong et al. [104] Launay et al. [165] Zhang et al. [313] Zhao et al. [314] Goff, [91] Repullo-Ruibérriz de Torres et al. [246]
Biology	• By increasing the diversity of weed species, it lessens the dominance of each weed and manages weeds. • Reduces the risk of yield loss and enhances the biodiversity of the cropping system.	Dokken et al. [72] Lin et al. [173] Murphy et al. [205] Aktar et al. [8] Gao et al. [86] Cheng et al. [53] Torralba et al. [285] Zohry et al. [317] Degani et al. [63] Wan and Wang, [296] Shabani et al. [257] Xie et al. [306] Brankov et al. [43]

8.3 Intercropping system

Research from drought-prone regions like Tanzania and Sweden highlights the resilience-enhancing benefits of diversified agricultural systems, including incorporating wild varieties, increasing temporal and spatial crop diversity, and promoting ecologically complex systems [278]. Similarly, after Hurricane Mitch, Nicaraguan farms with diverse cropping systems experienced lower erosive stress and economic losses, showcasing the advantages of structural complexity [102]. In southern Africa, crop diversification among smallholders has improved yields, incomes, food security, and nutrition [179]. Crop diversification is crucial for boosting yields, maintaining soil health, and balancing soil biodiversity while reducing the burden of pathogens [25, 93]. According to a meta-analysis by Beillouin et al., [30], diversification increased biodiversity by 24% and crop yields by 14%. Consideration of plant functional categories, such as nitrogen-fixing

legumes, C3 and C4 grasses, and non-legume species, is crucial for enhancing soil health and ecosystem services [26, 183, 294]. Challenges in implementing increased diversification include market access, irrigation availability, land size, rights, poverty, and policy interventions [81]. Economic policies often favor monoculture row crops over diverse systems, but changing climate dynamics may alter this trend. Additionally, the misconception that monoculture yields more biomass hinders diversification efforts.

9 Soil microbe-plant interactions and soil health

The thriving health of the plant rhizosphere is attributed to a diverse array of beneficial microorganisms, including rhizobacteria, mycorrhizal fungi, and others that support plant growth. Plant genotypes influence rhizosphere composition through root exudates, thereby attracting specific microbial species. Microorganisms coevolved with plants around 400 million years ago, transitioning from aquatic habitats to soil, establishing crucial soil-microbe-plant systems for nutrient cycling, maintaining soil fertility, and carbon sequestration [82].

Root exudates released by plants directly impact the root microbiome, providing carbon and energy to soil microorganisms and significantly altering soil physicochemical properties. These exudates also act as signaling molecules for soil microorganisms, including various plant hormones and defense chemicals, influencing plant-microbe interactions [123, 181, 189]. The mycorrhizal association between fungi and plant roots enhances nutrient absorption by increasing the root surface area [27, 141]. Agricultural land contamination with heavy metals from industrialization and urbanization has led to their biomagnification in ecosystems. Phytoremediation, combined with mycorrhizal fungi, offers a sustainable approach to remove heavy metals from soil, ensuring ecosystem health [175, 187, 192]. Microbes also play critical roles in soil functions like organic matter decomposition, nutrient cycling, and plant growth regulation [98]. Understanding the taxonomic structure of beneficial soil microorganisms is crucial for sustainable agriculture, and bioinformatics tools facilitate functional analysis by linking microbial community data with biological functions [82, 166, 214, 291].

10 Indicators for evaluating soil health in cropping systems

Farmers and stakeholders are increasingly embracing “soil health practices” to improve soil quality, recognizing their crucial role in achieving higher crop yields, protecting the environment, and mitigating climate change. These practices aim to ensure economic productivity and ecological sustainability, preserving or restoring the land’s integrity. Healthy soil underpins essential ecosystem functions, including supporting biological productivity, regulating air and water quality, and sustaining plant, animal, and human life.

Simple and cost-effective methods are used to assess physical indicators of soil health, including bulk density, porosity, aggregate stability, and soil texture. Hydrological processes such as runoff, aeration, erosion, infiltration rate, and water holding capacity are linked to these indicators [256]. Weak cohesion, low aeration and root density, high surface runoff, low water infiltration rates, and mechanization challenges are all signs of poor physical characteristics [67]. Indicators of chemical health, such as pH, CEC, organic matter content, and nutrient levels, are essential for determining how effectively soil retains toxic chemicals that impact the environment and plant development [146].

Table 7 Important soil health indicators at the farm scale. Source: [126, 261]

Physical indicator	Chemical indicator	Biological indicator
Permeability	Base saturation percentage	Soil organic carbon
Structural stability	Cation exchange capacity	Microbial biomass carbon
Bulk and particle density	Contaminant availability	Bacterial density/biomass
Clay mineralogy	Contaminant concentration	Fungal density/ biomass
Soil Color	Contaminant mobility	Actinomycetes density/ biomass
Soil Consistency	Contaminant presence	Soil respiration
Soil depth	Electrical conductivity	Acid phosphatase activity
Hydraulic conductivity	Exchangeable sodium percentage	Dehydrogenase activity
Oxygen diffusion rate	Nutrient cycling rates	Alkaline phosphatase
Particle size distribution	pH	Aryl-sulfatase activity
Penetration resistance	Soil nutrient availability	Urease activity
Pore conductivity	Soil nutrient content	Soil diversity index
Pore size distribution	Sodium adsorption ratio	Soil alteration index
Soil strength		Fatty acid analysis
Soil tilth		Nucleic acid analysis
Soil Structure		
Temperature		
Soil porosity		
Water storage capacity		

Biological indicators focus on natural processes, such as carbon fixation and transfer by plants through photosynthesis. The steady accumulation of organic matter from plants and animals enhances soil quality and helps sustain the natural cycles within ecosystems. Effective utilization of organic carbon and minerals relies on their recycling within the ecosystem [17]. To optimize soil conditions and enhance its physical, chemical, and biological qualities, a deeper understanding of the impacts of continuous cropping is necessary. Table 7 presents the different soil quality indicators.

11 Technological advances in soil health management

Technological innovations are transforming soil health management, offering sustainable and efficient solutions through biofertilizers, precision agriculture (PA), and advanced remote sensing technologies.

11.1 Biofertilizers and soil health

Biofertilizers are eco-friendly alternatives to chemical fertilizers that enhance soil fertility and plant growth by fixing nitrogen, solubilizing phosphorus, and decomposing organic matter. Microorganisms such as *Rhizobium*, *Azotobacter*, and *Azospirillum* increase nitrogen availability [217], while *Pseudomonas*, *Bacillus*, and *Aspergillus* solubilize phosphorus, thereby improving plant uptake [34]. Biofertilizers also produce phytohormones and enzymes that promote root development and suppress pathogens, reducing the need for pesticides [48, 217]. Their long-term application enhances soil structure, water retention, and microbial diversity [32], making them a vital component of sustainable agriculture [217]. The functional microbial-based bio-stimulants AMF and PGPR have been found to have favorable impacts on crops. AMFs and PGPR are involved in root colonization, boosting soil production and plant tolerance to abiotic challenges [132]. Similarly, Joel et al. [131] reported that co-application of AMF and engineered nanomaterials stimulates root colonization and abiotic stress tolerance in plants.

11.2 Precision agriculture and soil sensors

Precision agriculture leverages GPS, drones, and variable rate technology (VRT) for real-time, site-specific soil management. By integrating weather forecasts, soil data, and remote sensing, it enables targeted input application, improving nutrient use efficiency and reducing environmental impact [38, 226]. Soil sensors monitor parameters such as moisture, pH, and nutrient levels, enabling precise irrigation and fertilization [46, 87, 268]. These tools help minimize nutrient leaching and greenhouse gas emissions, while identifying soil erosion hotspots for targeted conservation [97, 198].

11.3 Remote sensing and unmanned aerial vehicle (UAV) technology

Remote sensing and UAVs enable large-scale, high-resolution monitoring of soil properties, including moisture, organic matter, and nutrient levels [87]. Real-time data enables timely interventions, such as irrigation adjustments based on moisture stress [316]. When integrated with a Geographic Information System (GIS), these technologies provide spatial insights into soil variability for more precise management [204]. UAVs provide high-resolution, multispectral imagery for assessing nutrient levels, erosion, and drainage, and enable 3D land surface modeling [56]. Together with PA, these tools support sustainable practices by enabling variable-rate irrigation and fertilization, thereby minimizing chemical use and preserving soil health [204].

12 Global initiatives and policies for sustainable soil management

International programs emphasize the importance of soil in supporting life and reducing climate change. Sustainable soil management is promoted through information exchange and capacity building, supported by initiatives such as the Global Soil Partnership (GSP) and the FAO's World Soil Charter. Whereas the UNCCD aims to address land degradation and desertification, the 4 per 1000 Initiative concentrates on raising soil organic carbon to combat global warming. Policies such as the EU Soil Directive and the Land Degradation Neutrality (LDN) framework guide nations in soil conservation, encouraging practices such as agroforestry, RA, and soil restoration. These efforts highlight the collective global responsibility for preserving soil health. Organizations like FAO and UNEP play pivotal roles in global soil health advocacy. FAO leads through initiatives such as the Global Soil Information System (GLOSIS) and Soil Health Card programs, while UNEP integrates soil conservation into biodiversity and climate policies. Case studies demonstrate regional successes, such as India's Soil Health Card Scheme, China's Grain for Green Program, and Brazil's no-till farming practices. These examples emphasize the need for policy-driven approaches supported by international collaboration. Together, they provide a blueprint for sustainable soil management that balances food security, environmental conservation, and climate resilience across diverse ecosystems.

13 Challenges and barriers to implementing sustainable soil management practices

13.1 Economic, social, and institutional constraints

In many degraded and rainfed areas, farmers often get trapped in cycles where poverty and land degradation reinforce each other [24, 89, 193, 255]. Low productivity leads to low income, which in turn limits farmers' capacity to invest in soil conservation [152,

201]. Their limited financial resources compel them to cultivate low-value crops, rely on traditional inputs, and refrain from adopting improved practices such as using high-quality seeds or installing irrigation systems. Poor access to credit, weak institutions, and poverty-induced migration further erode human and social capital, hindering awareness and uptake of technology. These systemic issues necessitate technological support and targeted investments to break the cycle of degradation and poverty.

13.2 Technological and infrastructure barriers

Soil degradation varies across locations, requiring tailored management interventions rather than a universal solution [114]. Many sustainable soil management approaches, such as watershed development and subsurface drainage, require community participation because of the interconnected hydrology. Adoption is also constrained by weak enabling infrastructure, including limited market access and a lack of extension services. Improving access to inputs and ensuring fair pricing through better market connections is key to increasing adoption and boosting productivity.

13.3 Policy, tenure, and governance issues

Tenure insecurity remains a major deterrent to conservation investment, especially by tenant farmers who see limited long-term returns [152, 200]. Heavy reliance on subsidies (e.g., 80–90% in watershed and drainage projects) discourages private investment, as farmers await public schemes. This calls for shifting toward credit-based soil restoration business models, with government agencies focusing on technical support, particularly in severely affected regions (e.g., subsurface drainage in sugarcane areas of Maharashtra). Despite the benefits of soil conservation for ecosystem services, payment for ecosystem services (PES) remains insufficiently embedded in planning processes, thereby restricting the economic incentives necessary for sustainable land management [155].

13.4 Knowledge gaps and research needs

Natural resource management (NRM) technologies often offer delayed but long-term benefits that farmers may not fully recognize. For instance, watershed interventions often take 5–7 years to fully deliver their benefits, and the ecosystem services they provide extend beyond individual farms [155]. Research should prioritize breaking the degradation–poverty nexus, now worsened by climate change impacts, through resilient, climate-smart technologies [153]. There is a need for inclusive, cost-effective, and scalable soil management strategies that engage smallholders through action research and participatory development. Policies must be tailored to socioeconomic realities, with a focus on integrating private investment and community participation alongside public support.

14 Conclusion and future thrust

This review demonstrates that soil health management is fundamental to sustaining agricultural productivity under accelerating climate change. Diverse soil biological communities, particularly AMF, PGPR, cyanobacteria, and other beneficial microbes, regulate nutrient cycling, carbon sequestration, soil aggregation, and root–soil interactions that underpin ecosystem functioning. Climate change–induced abiotic stresses such as drought, salinity, heat waves, nutrient imbalances, and soil degradation intensify

vulnerabilities in cropping systems, making biologically driven soil processes even more critical. Evidence from conservation agriculture, regenerative agriculture, crop diversification, reduced tillage, intercropping, and INM confirms that these practices significantly enhance microbial diversity, soil organic matter, water retention, and plant resilience. Policy interventions, institutional capacity, and advances in technologies such as precision agriculture, remote sensing, and biofertilizers further strengthen the potential for soil-centered climate resilience. Collectively, the synthesis highlights that restoring and maintaining soil health is foundational for achieving sustainable productivity, ecological stability, and food security in the face of escalating climatic risks.

Future research must focus on integrating soil biological processes more deeply into climate-resilient agricultural planning. Priority areas include (i) long-term, field-scale evaluation of AMF-PGPR consortia under multi-stress environments such as combined drought-heat, salinity-drought, and waterlogging scenarios; (ii) development of robust microbial formulations with greater shelf life, stress tolerance, and compatibility with major cropping systems; and (iii) advanced monitoring tools using soil sensors, UAVs, and GIS for real-time assessment of soil biological and physicochemical indicators. Strengthening policy frameworks to reward ecosystem services, especially carbon sequestration, soil restoration, and watershed conservation, will be essential for large-scale adoption. Greater emphasis is needed on climate-smart extension models, community-based soil restoration programs, and blended financial mechanisms that attract private sector participation. Finally, future soil health strategies must embrace a systems perspective, integrating microbial technologies, regenerative practices, and socio-institutional innovations to build resilient, productive, and sustainable agroecosystems capable of withstanding the uncertainties of climate change.

Author contributions
All authors contributed to the study conception and design. Neeshu Joshi and Dinesh Jinger initiated and designed the need of paper. Neeshu Joshi, Dinesh Jinger, Shourabh Joshi, Ekta Joshi, Jitendra Kumar Sharma, Venkatesh Paramesh, Vijay Singh Meena, Suresh Dhaterwal, Manju Verma, Samarpal Yadav, Devideen Yadav, B.L. Lakaria, R.A. Jat, and Nandha Kumar N., Manoj Kumar, M. Madhu have contributed equally to the preparation of the manuscript. All authors have revised or edited the manuscript carefully. All authors have read and approved the manuscript.

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